



Module Introduction Transcript

Hello all, welcome to the course and this is your faculty Dr. Saurabh, I will be walking through various different types of things that we will be learning in this course and also I will try to maximize your understanding by giving you most of the real life examples that we go through and because this entire course is about statistical understanding, statistical inference using different types of models and different types of methods, so anyways all of these methods are going to be very very vital for us to understand how things work, right, so without a further ado let us get started and let us get into the course.

There are various different types of modules in this course, this course is about statistical inference and modeling, first understand different types of probability and statistical methods, then we will get introduced to inference methods, hypothesis testing and so on and then we will get introduced to modeling techniques, so that we can how do we basically infer about the data and all of those things and we are able to get an idea about them and we also have created different types of material which will be accessible to all of you and these materials would be given to you as a consumption material on various different types of web URLs which we will be sharing and also through the handouts that we will be sharing with all of you, okay.

Now in this course we will be dealing with some sort of statistical methods as I have told you plus some probability things which you might have forgotten about, right and also we will try to club them, use them in different ways such that they becomes very naive very easily explorable topics for you to understand about the data itself, okay, so this course would build a strong principles for all of us to infer about data using statistics and probability, okay and we will also understand different types of methods that we have to quantify about uncertainties in life and various different situations of life either in businesses or whatever we do, okay and these are various different types of things that we will be going through starting from week 1, we will start with foundations of probability then we will get into random variables and their distributions, so I will introduce you with very basic definition of probability, I will introduce you with random variables what they are then we will move into distributions, then we will move into statistical inference estimation, hypothesis, remember I told you about there are various different types of things that we have to infer about the data, these are the things that I was talking about.

Then, we have analysis of variance, how do we measure if things are different or differently distributed, what are the methods that we have, so there are there would be a lot of statistical test that we will be learning through in this module, then we will be getting into some simple analysis or predictive models like simple linear regression and correlation, how do you correlate let us say height and weights are linearly associated or there is a correlation between them, right, so those would be the points



wherein we get the details of correlations and then we will move into week 6, wherein we will talk about multiple linear regression.

Now, I was just telling you about how two variables are related, let us say if you would want to learn about how multiple variables are related, let us say I would want to infer how sales of a company is related to the amount of money that they spend in TV, radio or newspaper promotions, right, so you have multiple variables involved that is why in this case we call this entire methodology as multiple linear regression, then we will move into discriminant analysis, how do discriminate data points from each other, so that is what we will be learning in that module, then we will go into principle component analysis which is all about dimension detection.

So let us say if we have too many fields to in the data, how do we work with that much amount of information, right, so we generally do not use all of that information which are redundant, think of images that you have, images are high dimensional images can have a lot of redundant pixels, now if you would have seen initially when we used to use WhatsApp and send images from one or let us say from our phones to some of our friends or families, our pixels of the images used to get reduced or the dimension of the image used to get reduced or the quality of the image used to get reduced, but the image was still visible, I mean the content of the image was still visible, this is what we will be learning in principle component analysis, how do we reduce the dimension in the data such that the effective content of the data does not get removed, but still we are able to see into the data or the image or whatever we have as a data, right, then we will move into cluster analysis wherein we will be grouping together similar data points together and discriminating between non-similar data points, then we will learn about time series, so let us say you know in last 40 years of time how different types of let us say population growth, in last 40 years of time how did the population in India grew in a time series fashion, now if you would want to extrapolate that information and predict for next 20 years you can do that, and the method that we are going to use there is time series analysis, then we will learn about non-parametric methods, a lot of, I will introduce you when we will be going through this module itself, and then we will summarize the course, within this course you would also be doing a lot of practical hands-on, when I say practical hands-on what I mean is we will be using python, we will be using real life datasets, we will be using R and we will do some sort of statistical analysis on those datasets to infer about it, to get to know about them better and in various different sense let us say if I have a data about videos that I watch every day on YouTube or Netflix, I can infer about how much amount of time that I spent and I can also infer about how I am using my various different subscriptions, if I am not viewing anything why I am paying for it, so let us say you might be using Spotify and you might want to see how have you been using Spotify for a very long period of time and you might want to optimize your premium subscription that you have bought, so in all of these cases the data that you have, I am not talking about any industrial data or any business data.



I am talking about the data that you yourself generate, so all of those things we will try making things very familiar whatever we have, whatever data that we generate ourselves and we will try to use that data and we will try to see how these simple methods like probability and statistics helps us to do or to have that realization how to use that data to improve on ourselves as well, so with that we will introduce you with the next module or the first module of the course and I will be going through it, so I welcome all of you again and I hope the learning that you would have in this course would be the kind of learning that you and I have had in during our childhoods and learning through examples through doing things, not only by learning through going through video or going through this video, so what we will be doing is we will provide you interactive environment and we will let you play within that environment, so that you can also learn in a gamified way, so I would give you enough introduction about each module when we will be going through it because this was just an introduction about the course itself, I just wanted to take some point of some amount of time to get to introduce you to the course first, okay.

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